

Visuomotor Cooperative Manipulation with Teams of Quadrotors

Teams of quadrotors could lift and transport objects in construction, agriculture, and people and goods transport, but current methods rely on motion capture, explicit knowledge of the payload, and fixed attachment points. In this project, teams of two or three quadrotors will cooperatively lift and manipulate the pose of arbitrary objects using only onboard sensing, combining large-scale pretrained diffusion policies, reinforcement learning, and control theory. The student will start with a single quadrotor manipulating an object with one gripper. The work takes place in the Mandralis Lab, University of Cambridge; visiting students from ETHZ and EPFL are welcome.

PROJECT	Master's thesis	STUDENT	TBD
SEMESTER	Jan–July 2027	HOST	University of Cambridge
START	January 2027	SUPERVISOR	Prof. Ioannis Mandralis
DEADLINE	July 2027	ADVISORS	—
CONTACT	imm38@cam.ac.uk		

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1 Introduction

Aerial robots that can grasp, lift and reposition objects would extend automation to tasks in construction, agriculture and logistics that ground robots cannot reach, and where the payload often exceeds what a single vehicle can carry [1, 2]. Teams of quadrotors can share such loads, and cooperative transport has been demonstrated in several forms, from multiple vehicles rigidly grasping a common object [3] to teams carrying cable-suspended payloads, including in agile flight [4, 5].

These demonstrations rely on assumptions that do not hold outside the laboratory. The states of the vehicles and of the payload are usually provided by a motion capture system; the mass, inertia and geometry of the payload are known in advance; and the payload is attached to the vehicles at fixed, pre-defined points. Work towards onboard sensing exists [6, 7], but it addresses the transport of a known payload rather than the grasping and reorientation of an object whose properties are not known beforehand.

In parallel, visuomotor policies trained on large demonstration datasets, such as diffusion policies and vision-language-action models, allow fixed-base manipulators to grasp and manipulate a wide range of objects from camera images alone [8, 9, 10]. Recent work has begun to transfer such policies to aerial manipulators by accounting for the dynamics of the flying platform when actions

are generated [11]. Reinforcement learning in simulation, followed by transfer to hardware, has produced quadrotor controllers that operate close to the physical limits of the vehicle [12, 13].

This thesis combines these directions. The student will first develop a single quadrotor with a gripper that grasps, lifts and reorients objects using only onboard sensing: a pretrained visuomotor policy provides end-effector targets from camera images, and a low-level controller, trained with reinforcement learning or derived from control theory, tracks them while compensating for the unknown payload. The approach will then be extended to teams of two or three quadrotors that jointly grasp an object and control its pose, selecting their own grasp points and coordinating through onboard perception rather than external tracking.

2 Objectives

This project aims to:

- 1. Onboard platform.** Build a quadrotor with a gripper and an onboard sensing and compute stack (visual-inertial odometry, an RGB-D camera and an embedded GPU) that operates without motion capture. The lab's existing quadrotors will serve as the starting point.
- 2. Simulation and data.** Develop a simulation environment for one or more quadrotors with grippers interacting with objects of varying shape, mass and inertia, suitable for massively parallel reinforcement learning and for generating demonstrations with a privileged expert controller.
- 3. Single-quadrotor visuomotor manipulation.** Fine-tune a large-scale pretrained visuomotor policy (e.g. a diffusion policy or a vision-language-action model) to grasp, lift and reorient previously unseen objects with a single quadrotor, and refine the closed-loop behaviour through reinforcement learning in simulation and continual learning on hardware.
- 4. Cooperative manipulation.** Extend the approach to two or three quadrotors that grasp a common object at self-selected points and control its position and orientation, using tools from control theory to regulate the internal forces between vehicles and to guarantee stable behaviour of the coupled system.
- 5. Experimental validation.** Quantify the performance of the onboard system against a baseline that uses motion capture and known payload parameters, in terms of pose-tracking error, success rate over a set of unseen objects, and robustness to disturbances.

3 Work breakdown

The project will be structured into the following work packages:

WP1 Literature review and platform

Review work on cooperative aerial manipulation, visuomotor policy learning and learning-based quadrotor control. Assemble and characterise the quadrotor-gripper platform with its onboard sensing and compute, and validate state estimation against motion capture.

WP2 Simulation and demonstration data

Implement the quadrotor-gripper-object system in a GPU-accelerated simulator with contact, and randomise object geometry, mass, inertia and friction. Generate demonstrations with a privileged expert (e.g. model predictive control with full state knowledge) and, where useful, by teleoperation on hardware.

WP3 Single-quadrotor visuomotor grasping

Fine-tune a pretrained visuomotor policy on the demonstration data so that it outputs end-effector targets from onboard images. Interface it with a low-level controller that tracks these targets, and demonstrate grasping, lifting and reorientation of a single object on hardware.

WP4 Policy refinement

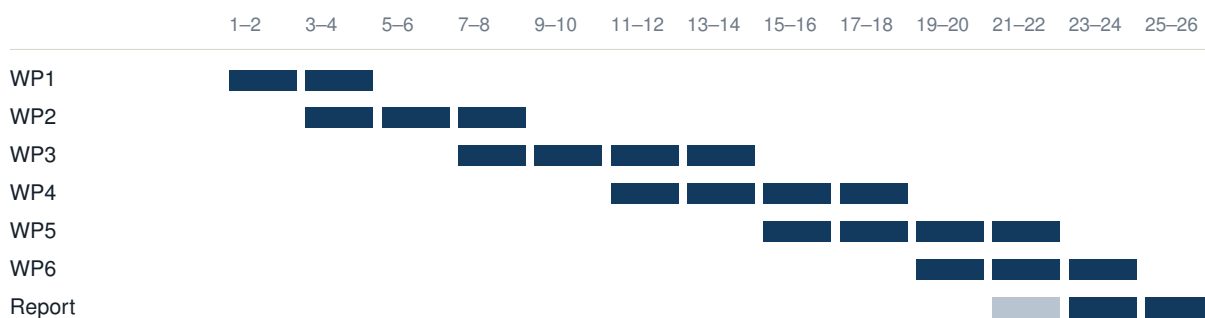
Train the low-level controller, or a residual on top of the pretrained policy, with reinforcement learning in simulation to compensate for unknown payloads. Investigate continual learning on hardware so that the policy improves from the robot's own experience.

WP5 Cooperative manipulation with two or three quadrotors

Extend the system to multiple vehicles grasping one object. Study decentralised policies that coordinate through onboard perception and limited communication, and use control-theoretic tools to regulate internal forces and ensure stability of the coupled system.

WP6 Experimental evaluation

Run an experimental campaign on a set of objects not seen in training and compare against a motion-capture baseline with known payload parameters. Report pose-tracking error, success rate and robustness to disturbances.

3.1 Indicative timeline**4 Candidate profile**

We are looking for a student with:

- A background in robotics, control or machine learning, from mechanical, electrical or computer engineering, computer science or a related programme.
- Programming experience in Python and PyTorch; experience with C++ and ROS 2 is an advantage.
- Familiarity with reinforcement learning, imitation learning or model-based control.
- Interest in experimental work; hands-on experience with drones or embedded hardware is an advantage.

To apply, send a CV, a transcript of records and a short statement of interest to the contact address above.

5 Deliverables and assessment

The student is expected to deliver:

1. A written report following the host university's guidelines, submitted by the deadline stated above.
2. A final presentation to the laboratory (20 min plus questions).
3. A documented code repository, comprising the simulation environment, the policy training and controller implementations, and the experimental data, structured so the work can be reproduced and carried on in follow-up projects.

Assessment follows the host university's regulations, with weight on the scientific quality of the work, the independence shown during the project, the quality of the documentation and the reproducibility of the results.

6 References

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Cambridge, TBD

SUPERVISOR

STUDENT

Prof. Ioannis Mandralis

University of Cambridge